



MIKROSKIL IDEAFUSE COMPETITION 2025

Oski and Kiki's Moving Mystery

Oski is teaching his nephew Kiki about patterns and sequences. To make it fun, Oski gives Kiki a challenge.

He hides a **sequence of numbers** A of length $N + 2$, then tells Kiki only the **moving sums of 3 consecutive elements**:

$$S_i = A_i + A_{i+1} + A_{i+2} \text{ for all } (1 \leq i \leq N).$$

Now Kiki wants to **reconstruct the original sequence** A .

Given the list of N sums ($S_1, S_2, \dots, S_N$), reconstruct one possible **integer sequence A of length $N + 2$** , such that:

$$S_i = A_i + A_{i+1} + A_{i+2}$$

for every valid i .

Among all possible valid sequences, output the one that is **lexicographically largest** (descending order comparison).

Input Format

```
T
N
S1 S2 ... SN
```

- $1 \leq T \leq 10$ -- denotes the number of test cases
- $1 \leq N \leq 300,000$ -- denotes the number of sums
- $3 \leq S_i \leq 6,000$ -- denote sum of each number where ($1 \leq A_i \leq 2,000$)

Output Format

For each test case:

```
Case #x: A1 A2 ... A(N+2)
```

The sequence must satisfy all conditions and be **lexicographically largest**.

Sample Input

```
1
4
5 8 7 8
```

Sample Output

```
Case #1: 2 2 1 5 1 2
```

Explanation

$$2+2+1 = 5$$

$$2+1+5 = 8$$

$$1+5+1 = 7$$

$$5+1+2 = 8$$

all match (5,8,7,8)

Other valid sequences are 1, 3, 1, 4, 2, 2, 1, 2, 2, 4, 1, 3. Among all valid sequences, this 2, 2, 1, 5, 1, 2 is lexicographically largest.